Ali Ragab Ghonim

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PROFESSIONAL SUMMARY

Backend Software Engineer specializing in Java and Spring Boot, with experience building scalable REST APIs, AI-powered applications, Retrieval-Augmented Generation (RAG) pipelines, and LLM-based agent workflows. Passionate about designing clean, production-ready backend systems using Clean Architecture, Spring Security, PostgreSQL, Docker, and modern cloud-native practices. Graduate of ITI's Open Source Track (9-month intensive program).

TECHNICAL SKILLS

Languages & Frameworks: Java 17, Spring Boot 3.x, Spring Security, Spring Data JPA, Hibernate, Spring AI, NestJS, TypeScript, PHP (Laravel), C# (ASP.NET Core)

Testing: JUnit 5, Mockito, Unit Testing, Integration Testing, Test-Driven Development (TDD)

Databases: PostgreSQL, MySQL, MongoDB, MongoDB Vector Search, Redis, Hibernate ORM

Backend & AI: REST API Design, Clean Architecture, RAG Pipelines, LLM Agents, OOP, SOLID, JWT, OAuth2, RBAC

DevOps & Tools: Docker, Git, GitHub, Linux, CI/CD, BullMQ, Postman, Swagger, OpenAPI 3, Agile, Scrum

Frontend (Familiar): React, Angular, Bootstrap 5

PROJECTS & EXPERIENCE

CoursePilot — Full-Stack E-Learning Platform — Full-Stack Engineer

Personal Project | Java / Spring Boot / React | 2026 | GitHub Repository

- Built an AI-powered course assistant using Spring AI, integrated with a React frontend for real-time student interaction.
- Designed and implemented a REST API with Spring Boot 3.x following Clean Architecture; secured endpoints using Spring Security, JWT, and role-based access control (Admin, Instructor, Student).
- Built course lifecycle management (Draft / Published / Archived) with full Swagger / OpenAPI documentation, achieving high test coverage with JUnit 5 and Mockito, and containerized the full stack using Docker.

Tech Stack: *Java 17, Spring Boot 3.x, Spring AI, Spring Security, JWT, React, PostgreSQL, Redis, JUnit 5, Mockito, Docker*

StockSavvy — AI Inventory Management Platform — Full-Stack Engineer

ITI Graduation Project | NestJS / TypeScript / React | 2026 | GitHub Repository

- Developed an AI-powered inventory management platform using Retrieval-Augmented Generation (RAG) and LLM-powered autonomous agents to assist purchasing, inventory monitoring, and supplier negotiations.
- Engineered a multi-agent architecture (Demand Forecasting, Reorder Automation, Vendor Negotiation, Anomaly Detection) with human-in-the-loop approvals, audit logging, and async workflows using BullMQ and Redis.
- Built a Retrieval-Augmented Generation (RAG) pipeline using PostgreSQL (pgvector) to perform semantic search over vendor contracts, pricing history, and negotiation transcripts, enabling grounded, context-aware AI responses.

Tech Stack: *NestJS, TypeScript, React,* PostgreSQL , pgvector*, Redis, BullMQ, Docker, OpenAI API*

EDUCATION

Information Technology Institute (ITI) — Open Source Track

9-Month Intensive Program | Shebin El Kom, Egypt | Oct 2025 - Jul 2026

Java EE, Angular, Linux, Database Design, Agile/Scrum, DevOps; multiple full-stack projects with code reviews and team collaboration.

B.Sc. Mathematics & Computer Science — Minufia University

Shebin El Kom, Egypt | Sep 2021 - Jun 2025 | GPA: 3.3 / 4.0

Data Structures, Algorithms, OOP, Database Systems, Software Engineering, Discrete Mathematics.

TRAINING & CERTIFICATIONS

ITI Summer Training — .NET Development Track

Information Technology Institute | Jul 2023 - Sep 2023

120-hour C# and ASP.NET Core training covering MVC, Entity Framework Core, and RESTful API development; delivered task management app as capstone project.

LANGUAGES

Arabic (Native) | English (Professional Working Proficiency, B2+)